

Libraries, spaCy & Tokens

Pre-Assessment 1 · 11 questions · 38 points

Name: _____

Date: _____

Score: / 38

This is a **pre-assessment**. It shows what you already know **before** we start this unit, so it is fine if some questions cover things you have not learned yet. Try every question you can. If you have no idea at all, write “**not yet**” and move on.

Work on your own, without a computer. You have about **45 minutes**. When a question asks what a program prints, write exactly what Python would print, one line of output per answer line. If the code would crash, write the name of the error (for example, `TypeError`). Show your thinking; partial credit is given.

In every spaCy question, assume this code has already run: `import spacy` and `nlp = spacy.load("en_core_web_sm")`

1. What is a library, really?

4 points

Meanings:

1. A single `.py` file of code you can import.
2. A folder of modules grouped together and installed as one unit.
3. A big collection of code for one kind of job, like spaCy.
4. The set of functions and classes a library lets you use, and the rules for calling them.

(a) Which meaning number matches **module**?

(1 pt)

(b) Which meaning number matches **API**?

(1 pt)

(c) In `from random import randint`, what is `random`?

(1 pt)

(d) In `from random import randint`, what is `randint`?

(1 pt)

2. Installing libraries safely

3 points

(a) Write the command that creates a virtual environment in a folder named `.venv`. (1 pt)

(b) Write the pip command that shows details (version, location) about the installed `spacy` package. (1 pt)

(c) What is a `requirements.txt` file for? (1 pt)

3. Why language is hard

2 points

Read the sentence: **"The chicken is ready to eat."**

Describe **two** different meanings this sentence could have.

4. Hello, spaCy

3 points

```
doc = nlp("Pandas eat bamboo.")
print(len(doc))
print(doc[-1].text)
```

(a) What kind of object does `nlp(...)` give back? (1 pt)

(b) What does `print(len(doc))` show? (1 pt)

(c) What does `print(doc[-1].text)` show? (1 pt)

5. Tokenization: beyond split()

4 points

```
text = "I don't like 5.5 km walks!"  
print(len(text.split()))  
print([t.text for t in nlp(text)])
```

(a) What does the first `print` show?

(1 pt)

(b) Write spaCy's tokens in order, separated by `|`.

(2 pts)

(c) Name one thing spaCy's tokenizer handles better than `.split()` in this sentence.

(1 pt)

6. Blank or trained?

2 points

(a) You only need to split thousands of texts into tokens, as fast as possible. Which line is best? (1 pt)

(A) `nlp = spacy.blank("en")`

(B) `nlp = spacy.load("en_core_web_sm")`

(b) Would `token.pos_` give useful part-of-speech labels with `spacy.blank("en")`? Why or why not?

(1 pt)

7. Token Sorter

5 points

Write **T** (True) or **F** (False) in every box. Each row is one token.

Token	is_alpha	is_digit	like_num	is_punct	like_url
Maya					
42					
ten					
?					
kidcode.org					

8. Doc indexing

3 points

`doc = nlp("The small robot cleaned my room.")`. Write what each expression gives.

Expression	Result
<code>doc[2].text</code>	
<code>doc[1:3].text</code>	
<code>doc[-2].text</code>	

9. Clean-Up Crew

4 points

Fill in the blanks so `clean` keeps the **lowercase** text of every token that is **not** a stop word and **not** punctuation. Then write what it prints.

```
doc = nlp("The Dogs and the cats ARE playing!")
clean = [t._____ for t in doc if not t._____ and not t._____]
print(clean)
```

(a) Write the three missing attribute names, in order.

(3 pts)

(b) What does it print?

(1 pt)

10. Word Family Finder

3 points

spaCy reads: "The boy was running while the mice ate cheese."

(a) Write the `lemma_` of each word.

(2 pts)

Word	running	mice	ate	was
<code>lemma_</code>				

(b) Why does counting lemmas instead of words give fairer word counts?

(1 pt)

11. Cleaning recipe

5 points

Write a function `clean_lemmas(text)` that runs `nlp` on the text and uses a **list comprehension** to **return** the lowercase lemma of every token that is not a stop word and not punctuation.

EXAMPLE: `CLEAN_LEMMAS("THE CATS WERE RUNNING!")` RETURNS

```
['cat', 'run']
```

END OF ASSESSMENT